CreditPathAI – Milestone 3: Baseline Model Development

1. Objective

The objective of this milestone was to build a **baseline machine learning model** using Logistic Regression to predict loan default risk and evaluate its performance using the **AUC-ROC metric**.

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2. Dataset Used

The model was trained on the **feature-engineered dataset (`final_features.csv`)**, which was created after data cleaning and feature engineering.

- Raw dataset → cleaned dataset → engineered features
- Target variable: `default` (0 = No Default, 1 = Default)

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3. Feature Engineering Summary

The following important features were created:

- **Loan-to-Income Ratio** → Measures financial burden
- **Credit Utilization** → Debt per credit line
- **Interest Burden** → Loan × interest rate
- **Employment Stability** → Job duration vs age
- **High DTI Flag** → Indicates high risk borrowers
- **Credit Score Bucketing** → Categorized credit levels
- **Log Income** → Handles skewed income distribution
- **Interaction Feature** → Income × Loan amount

These features help the model better understand borrower risk patterns.

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4. Model Selection

A **Logistic Regression model** was used as a baseline because:

- It is simple and interpretable
- Works well for binary classification

- Provides feature importance via coefficients

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5. Data Preprocessing

- Data was split into **80% training and 20% testing**
- **StandardScaler** was applied (mandatory)
- Class imbalance handled using:

class_weight = 'balanced'

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6. Model Training

The Logistic Regression model was trained on scaled training data with:

- Max iterations = 1000
- Balanced class weights

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7. Evaluation Metric

The model was evaluated using **AUC-ROC (Area Under Curve)**.

👉 **Why AUC-ROC?**

- Measures model's ability to distinguish between classes
- Better than accuracy for imbalanced datasets

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8. Results

- **AUC-ROC Score:** ___ *(fill your value)*

Interpretation:

- 0.70 → Good

- 0.75 → Very Good

- 0.80 → Excellent

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9. ROC Curve Analysis

- ROC curve plotted between True Positive Rate and False Positive Rate
- Curve closer to top-left indicates better performance
- Graph saved as: `roc_curve.png`

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10. Confusion Matrix

- Shows actual vs predicted classification
- Helps understand model errors
- Saved as: `confusion_matrix.png`

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11. Feature Importance (Model Interpretation)

🔺 **Features Increasing Default Risk:**

- loan_income_ratio
- high_dti_flag
- interest_burden

🔻 **Features Reducing Default Risk:**

- employment_stability
- log_income

Positive coefficients → increase risk
Negative coefficients → reduce risk

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12. Conclusion

The Logistic Regression model achieved a **moderate to strong performance** and successfully captured important financial risk patterns.

This model serves as a **baseline benchmark** for future improvements using advanced models like XGBoost.